

Stable Matching Problem

input:

n hospitals

$$H = \{h_1, h_2, \dots, h_n\}$$

n students

$$S = \{s_1, s_2, \dots, s_n\}$$

$$|H| = |S| = n \quad \leftarrow$$

$\uparrow$
cardinality
size

preference lists for each hospital and each student

n^2 for students

n^2 for hospitals

output:

perfect matching that is self-reinforcing stable

set builder notation, set abstraction

$$\text{A set } M \subseteq H \times S = \{ (h, s) : h \in H \text{ and } s \in S \}$$

$\uparrow$
subset
of

$\uparrow$
cartesian
product

$\uparrow$
element
of
in

is a perfect matching if

- ① each $h \in H$ is in at most one pair of M
- ② each $s \in S$ is in at most one pair of M
- ③ $|M| = |H| = |S| = n$

example

$$M = \{(A, X), (B, Z), (C, Y)\}$$

$A: \textcircled{X} Y Z$
 $B: Y \textcircled{X} \textcircled{Z}$
 $C: X \textcircled{Y} Z$
 $\uparrow$
 hospitals

$X: \textcircled{B} \textcircled{A} C$
 $Y: A B \textcircled{C}$
 $Z: A \textcircled{B} C$
 $\uparrow$
 students

A perfect matching M is stable if it contains no unstable pairs.

Given M , $h \in H$ and $s \in S$ form an unstable pair if:

- ① $(h, s) \notin M$
 $\uparrow$
 not an element of M
- ② h prefers s over h 's current match
- ③ s prefers h over s 's current match

Is M stable? $M = \{(A, X), (B, Z), (C, Y)\}$

Does M have any unstable pairs?

(A, X) an unstable pair? NO - fails ①, $(A, X) \in M$

(B, X) an unstable pair?

① $(B, X) \notin M$ ^{yes} ② B prefer X over Z ^{yes}

③ Does X prefer B over A ? ^{yes}

Is there a stable matching?

$A: X \text{ } \textcircled{Y} \text{ } Z$

$B: Y \text{ } \textcircled{X} \text{ } Z$

$C: X \text{ } Y \text{ } \textcircled{Z}$

$X: \textcircled{B} \text{ } A \text{ } C$

$Y: \textcircled{A} \text{ } B \text{ } C$

$Z: A \text{ } B \text{ } \textcircled{C}$

proposed $M = \{(A, Y), (B, X), (C, Z)\}$

How to check that M is stable?